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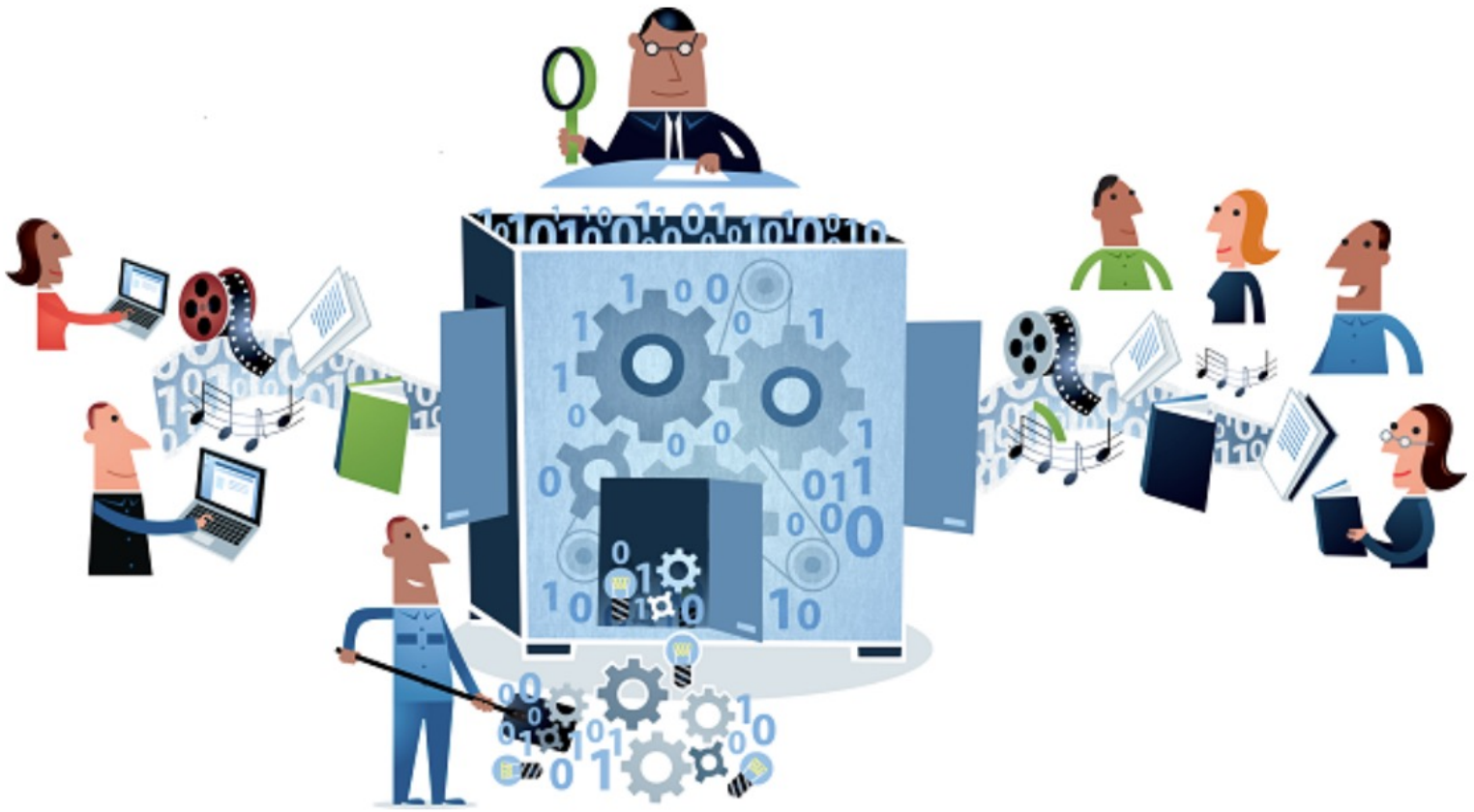

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Introduction to Archives and Digital Preservation

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Agenda

- Definition of ‘Archives’
- The role of public archives in society
- Preservation of digital archives

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What are archives?

- Archives are collections of items or ‘records’ which have been selected for permanent preservation because
  - of their value as evidence
  - or as a source for historical or other research.

- Records are created by the activities of organisations and people; they serve an active purpose whilst in current use and some of them are later selected and preserved as part of an archival collection.

- Archive collections are usually unique, which is why it is so important to take proper care of them.

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https://upload.wikimedia.org/wikipedia/commons/b/b4/Archives_nationales_%28Paris%29_la_galerie_du_Parlement_%28Grands_D%C3%A9p%C3%B4ts%29.png

Archives as institutions

- Often these collections are kept in specialized institutions, which are also called ‘archives’.
- Examples of these include national and cantonal archives, but also archives that are specialized for certain topics.
- Archives are also kept by other institutions, including museums and libraries.

Swiss Federal Archives – Schweizerisches Bundesarchiv
https://www.bar.admin.ch/bar/de/home/ueber-uns/das-bundesarchiv.html

The role of archives in society

- Archives allow a time-delayed traceability and monitoring of the activities of administrations and governments. In this way, they have an important control function in a democracy.

- Provide evidence of activities which occurred in the past, they tell stories, document people and identity and are valuable sources of information for historic research.

- They are our recorded memory and form an important part of our community, cultural, official and unofficial history

- Guarantee legal reliability

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«Work» vs. «Record»

A rough distinction between libraries and archives:
Libraries contain works, archives contain records.

Works in the sense of copyright law
- Independent artistic creations such as works of art, literature, music and paintings.
- A work is a protected or protectable creation within the meaning of copyright law.

Records
- Output of administrative processes in the form of documents or files
- Are kept in “records management systems”
- Make activities and decision making processes traceable (economic activities, administrative activities)

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Two main principles for structuring archives

Provenance

Provenance refers to the individual, family, or organization that created or received the items in a collection. The principle of provenance or the **respect des fonds** dictates that records of different origins (provenance) be kept separate to preserve their context.

Archives are not structured according to content-related topics, but according to provenance

Original Order

Archives are kept in the order in which they were originally created or used.

It is essential to understand this when working with archives so that this original order is preserved. This original order allows custodians to protect the authenticity of the records and provides essential information as to how they were created, kept and used.

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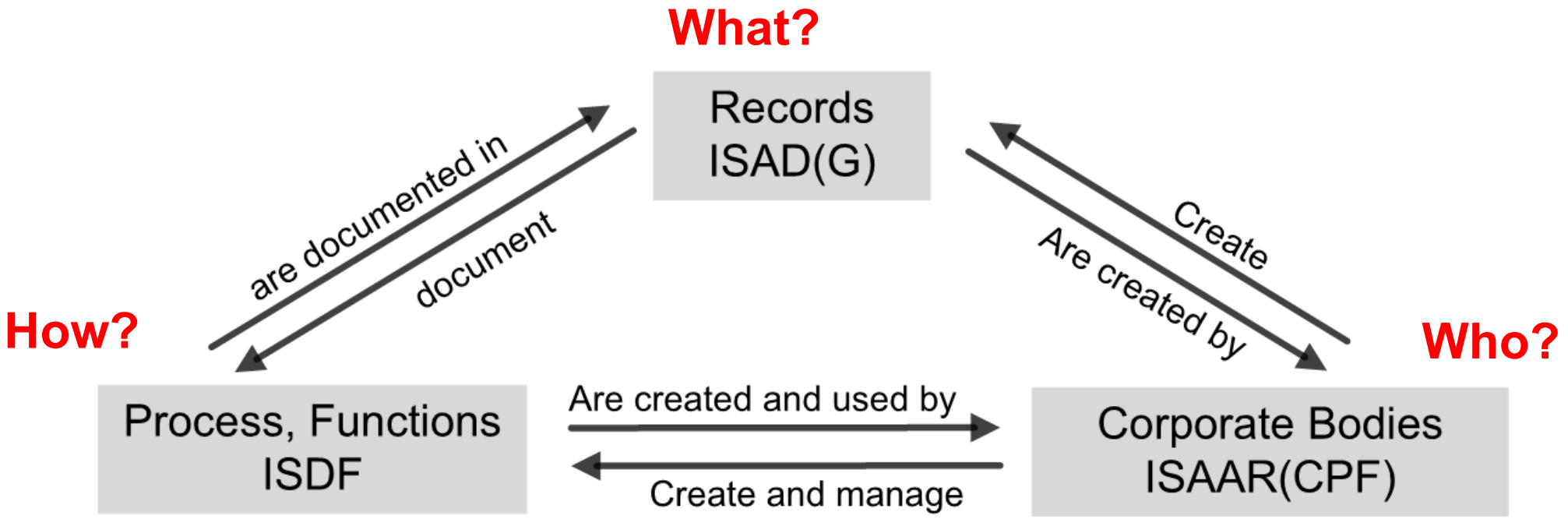
https://dictionary.archivists.org/entry/provenance.html
https://dictionary.archivists.org/entry/original-order.html

How to describe and contextualise archival material?
The status quo in archives

- The content of archives is currently described in **ISAD(G), "International Standard on Archival Description"**
- The archival standards ISAD(G), ISDF and ISAAR(CPF) have been developped by the International Council on Archives (ICA, https://www.ica.org/)
- ISAD(G) models the content and context of archival material in a hierarchical manner.
- Exactly one hierarchy can be used, thus it's a monohierarchical model of description.



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Confederaziun svizra

Online access to the Federal Archives

1.6.0.1077

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Archive plan

Archive plan

- **CH-BAR#Unterlagen*** Gesamtbestand des Schweizerischen Bundesarchivs (1798-) (1424 - 2007)
  - **B*** Archiv der Helvetik (1798-1803) (1712 - 1857)
  - **C*** Archiv der Mediationszeit (1803-1814) (1685 - 1843)
  - **D*** Archiv der Tagsatzungsperiode (1814-1848) (1715 - 1860)
  - **E*** Archiv des Schweizerischen Bundesstaates ab 1848 (1848 - 2018)
    - **E1*** Bundeskanzlei (1828 - 1970)
      - **E1#1** Bundeskanzlei (1848 - 1951)
      - **E1#2** Bundesrat (1848 - 1951)
        - **E1#2.1** Reden, Ansprachen, Appelle (1848 - 1950)
          - **E1#2.1.1** Celio, Enrico (1848 - 1950)
            - **E1#1000/4#268*** Politik und Technik im Dienste des Landes. Ansprache anlässlich der Schweizerische…(01.01.1943 - 31.12.1943)
            - **E1#1000/4#270*** Frei und neutral bleiben. Ansprache anlässlich der 1. August-Feier in Locarno (01.01.1945 - 31.12.1945)
            - **E1#1000/4#271*** Die wirtschaftliche und politische Bedeutung der schweizerischen Elektrizität. Ans… (01.01.1945 - 31.12.1945)
            - **E1#1000/4#276*** Ansprache am internationalen Eisenbahnkongress in Luzern. Juni 1947 (01.01.1947 - 31.12.1947)
            - **E1#1000/4#277*** 1847-1848: Zwei eng verwandte Daten. Ansprache anlässlich der Jahrhundertfeier … (01.01.1947 - 31.12.1947)
            - **E1#1000/4#278*** Ansprache gehalten in Crans s/Sierre über die grossen Pioniere des Walliser Fremd… (01.01.1947 - 31.12.1947)

ISAD(G):
Contextualisation + description within a hierarchical model

Swiss Federal Archives
https://www.recherche.bar.admin.ch/recherche/#/en/search/archive-plan

The goals of digital preservation

- Keeping digital information usable in the long term
  → **"long term" =** across multiple generations of hardware, operating systems, software / applications and file formats.
  → **"usable" =** find, display, interpret, reuse.

- The digital archive documents the context of origin of records and ensures the long-term traceability of administrative business processes.

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The basic challenge of digital preservation

- No direct link between data producer and archives user.
- What additional information (metadata) is needed to ensure that the packaged information remains usable?

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Digital archiving in 5 steps

1. Appraisal, selection
2. Preparation and ingest
3. Storage
4. Preservation planning
5. Access

Appraisal, selection

- **Why** should we archive?
  Mandate, archiving strategy, legal basis, historic value
- **What** should be archived?
  Evaluation and selection of documents and metadata.
- **For whom** should the archives be preserved?
  Designated community, community of stakeholders and users that the archives serve.
  - Digital archival content does not have to be understood and interpreted by everyone in the future.
  - However, it must be possible for the designated community to interpret the information based on their expert knowledge (without a producer).

- Only **3-5%** of the records and data produced in a public administration wil be preserved in an archive.

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Preparation of data for archiving

Preparation of data and metadata in automated processes

- Virus check
- Quality control and validation against file format specifications
- File migration to archival formats if needed
- Extracting technical and descrive metadata from the files
- Assign unique identifiers
- Storage of the files and their metadata in standardized form in repositories
- Update the archive catalogue with the descriptive metadata
- Documentation of all steps. The documentation is an integral part of the archiving process and ensures traceability.

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Archival storage

Management of the storage (repository)

- Repository = Software for managing fixed digital information
- Interface for receiving and delivering data
- Automated integrity checks, monitoring and alerting in case of errors
- User management

Bitstream preservation

- Storage system (disks, tapes etc.)
- Protection of the bitstream through redundancy and diversity
- General requirement: 3 copies in 2 geographical locations, at least one copy offline
- Fixity (checksum), monitoring, repair if necessary

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Preservation planning

- Preservation planning is the function for monitoring changes that may impact on the digital material that the repository holds.
- Preservation planning is proactive

- **Technology watch**
  - File formats
  - Storage technologies
  - Tools and environment
  - Access mechanisms
- **Designated communities**
  - Needs and expectations of users
  - Needs and expectations of producers
  - Emerging tools for machine to machine access

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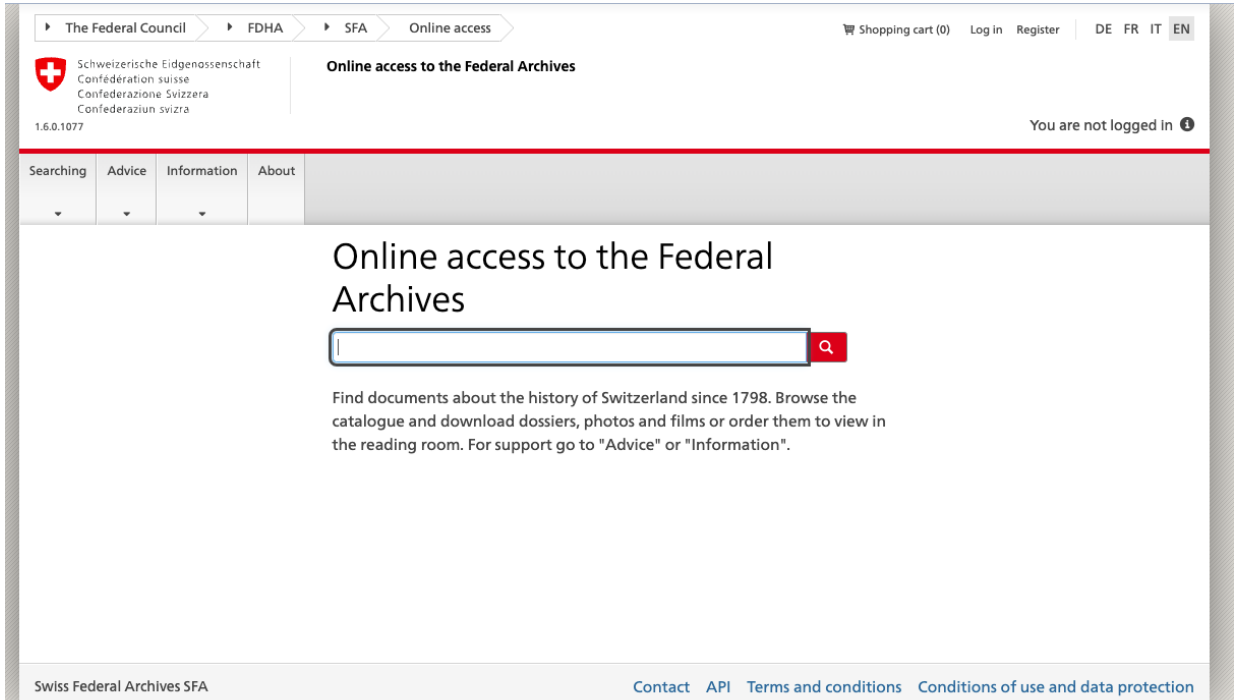
Access, use, reuse of the archives

- Understanding user needs
- Mapping archival formats (TIFF, no OCR etc.) with user needs (instant web access)
- Contradiction between efficient access and data protection, lockup periods
- Access
  - Digital reading rooms, normally built as web applications for human beings
  - Interfaces for machines, API, SPARQL endpoints in the case of semantic data



https://www.recherche.bar.admin.ch/recherche/

Conclusion and main points

- Archives have the duty to **make government activities traceable.**
- The two main principles for structuring archives: **provenance** and the **original order**
- Archival description is based on **ISAD(G), «International Standard on Archival Description»**
- The basic challenge of digital preservation: We need the right metadata to **contextualise** digital information and **keep it usable** in the long term.

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Thank you for your attention.

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Illustrations: Jørgen Stamp, https://digitalbevaring.dk/

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